which he called *the profile tracer*. It looked something like a lawnmower with two bicycle wheels and a pendulum, and as was pushed over land, integrating and recording horizontal and vertical measurements. Vannevar thought it would be commercially successful, but it never caught on. The young man learned many things from this failure—to become a real engineer he needed to learn more than math and physics. He needed to learn how to effectively deal with people.

The same difficulties faced Bush during the WWI, when he was working for US Navy, to which he proposed his idea for a device, that would use magnetic fields to detect submarines. His device proved to be successful in testing, but Navy officials, did not deploy the device correctly and it proved virtually useless in combat. Bush again learned that a successful engineer also had to be a good politician.

In 1919, Bush left Tufts and went to MIT's electrical engineering department as a professor of electrical power transmission. Starting from 1925, Bush began construction of mechanical devices for performing integration. First was the *Product Intergraph*, next was the *Differential Analyzer*, an analog computer that could solve differential equations with as many as 18 independent variables. It actually used large gears and other mechanical parts to solve equations. In 1931, he completed the first differential analyzer (see the lower photo). At the same tine, Bush and team at MIT develop the "network analyzer", a system for setting up miniature versions of large and important electrical networks. Later Bush constructed an improved *Differential Analyzer*, built by electronic tubes, which was used intensively during the WWII.

Bush and his *Differential Analyzer*

In 1922, Bush, his college roommate, Laurence Marshall, and a colleague of Bush from the company Amrad (a small company for radio devices, where Vannevar served as a consultant), Charles G. Smith, set up the *Metals and Controls Corporation*, for producing thermostats and thermionic tubes. Later the company managed to market successfully a device, called the *S-tube*. This was a gaseous rectifier, invented by Charles Smith (the patent was purchased by Amrad), that greatly improved the efficiency of radios, eliminating the need for radio batteries. Bush made much money from the venture. The company, renamed *Raytheon* in 1925, became a large electronics company and defense contractor, and still exist now as a major American defense contractor and industrial corporation.

Bush also worked on developing machines that would automate human thinking. Specialization in just about every field of science was creating a glut of information. Something was needed to help sort through the growing store of accumulated knowledge. In the 1930s microfilm, which had been around for some decades, was growing in popularity as a storage device, especially among librarians. Bush, who was a photography enthusiast, was quite interested in this resurgent technology. He proposed to build a machine for the FBI that could review 1000 fingerprints a minute. They however turned him down. In 1938 Bush and John Howard built and patented *the rapid selector*, a machine designed for high-speed referencing of information, stored on microfilm. In